THIRUVENKADAPRASATH.S

192312016

12	Implement Iris Flower Classification using KNN

CSV FILE

```
SepalLength,SepalWidth,PetalLength,PetalWidth,Species
5.1,3.5,1.4,0.2,Setosa
4.9,3.0,1.4,0.2,Setosa
4.7,3.2,1.3,0.2,Setosa
7.0,3.2,4.7,1.4,Versicolor
6.4,3.2,4.5,1.5,Versicolor
6.9,3.1,4.9,1.5,Versicolor
6.3,3.3,6.0,2.5,Virginica
5.8,2.7,5.1,1.9,Virginica
7.1,3.0,5.9,2.1,Virginica
```

PROGRAM

```
import numpy as np
import pandas as pd



class KNNScratch:

    def __init__(self, k=3):
        self.k = k
        self.X_train = None
        self.y_train = None

    def fit(self, X, y):
```

```
        self.X_train = X
        self.y_train = y

    def _euclidean_distance(self, x1, x2):
        return np.sqrt(np.sum((x1 - x2) ** 2))

    def _predict_sample(self, x):
        distances = [
            self._euclidean_distance(x, x_train) for x_train in self.X_train
        ]
        k_indices = np.argsort(distances)[: self.k]
        k_nearest_labels = [self.y_train[i] for i in k_indices]

        labels, counts = np.unique(k_nearest_labels, return_counts=True)
        return labels[np.argmax(counts)]

    def predict(self, X):
        return np.array([self._predict_sample(x) for x in X])


df = pd.read_csv("iris_knn_data.csv")

X = df[
    ["SepalLength", "SepalWidth", "PetalLength", "PetalWidth"]
].values
y = df["Species"].values

model = KNNScratch(k=3)
model.fit(X, y)
```

```
predictions = model.predict(X)

accuracy = np.mean(predictions == y) * 100

print(f"Model Training Accuracy: {accuracy:.2f}%")

sample = np.array([[6.5, 3.0, 5.2, 2.0]])
predicted_species = model.predict(sample)[0]

print(
    f"Predicted Species for sample [6.5, 3.0, 5.2, 2.0]: {predicted_species}"
)
```

OUTPUT